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Systems and Methods for Ephemeral Processing of High Cardinality Data

Abstract

Systems and methods for ephemeral processing of high cardinality data. The system can receive log data indicative of metrics associated with a computing system, wherein the log data is received by an aggregation layer. The method includes aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to increase or decrease a level of deduplication. The method includes, in response to aggregating the log data, determining deduplicated log data including one or more unique logs indicative of unique metrics associated with the computing system. The method includes transmitting the deduplicated log data to a storage system.

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Background/Summary

FIELD

[0001] The present disclosure generally relates to techniques for processing data received by a computing system.

BACKGROUND

[0002] Log data may include various types of information associated with a computing system. For instance, log data may record all events occurring within a computing system, an application, or on a network device. Logging may be enabled to facilitate the monitoring and analysis of log data. However, processing and storing large quantities of logs may increase storage costs and consume considerable computing resources adversely impacting the computing system.

SUMMARY

[0003] Aspects and advantages of embodiments of the present disclosure will be set forth in part in the following description, or may be learned from the description, or may be learned through practice of the embodiments.

[0004] In an example aspect, the present disclosure provides an example computer-implemented method. The example computer-implemented method includes receiving, from one or more client devices, log data including a plurality of logs indicative of metrics associated with a computing system, wherein the log data is received by an aggregation layer. The method includes aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to increase or decrease a level of deduplication. The method includes, in response to aggregating the log data, determining deduplicated log data including one or more unique logs indicative of unique metrics associated with the computing system. The method includes transmitting the deduplicated log data to a storage system.

[0005] In some implementations, the log data includes at least one of: (i) a higher throughput than the deduplicated log data or (ii) a higher cardinality than the deduplicated log data.

[0006] In some implementations, the aggregation layer is provided as part of a network layer.

[0007] In some implementations, the log data is ephemerally received by the aggregation layer.

[0008] In some implementations, the method includes accessing the deduplicated data from the storage system. In some implementations, the method includes computing analytics data based on the deduplicated data, the analytics data being associated with the log data.

[0009] In some implementations, the one or more aggregation parameters includes metadata associated with the plurality of logs.

[0010] In some implementations, the metadata includes at least one of: (i) a time stamp or (ii) a log type associated with respective logs of the plurality of logs.

[0011] In some implementations, the method includes determining the aggregation layer is unavailable to receive additional log data. In some implementations, the method includes, based on determining the aggregation layer is unavailable to receive the additional log data, transmitting the additional log data to the storage system.

[0012] In some implementations, determining the aggregation layer is unavailable to receive the additional log data includes determining one or more faults associated with the aggregation layer, wherein the one or more faults are indicative of a fault tolerance.

[0013] In another aspect, the present disclosure provides an example computing system. The example computing system includes one or more processors and one or more non-transitory, computer readable medium storing instructions that are executable by the one or more processors to cause the computing system to perform operations. The example operations include receiving, from one or more client devices, log data including a plurality of logs indicative of metrics associated with a computing system, wherein the log data is received by an aggregation layer. The example operations include aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to

increase or decrease a level of deduplication. The example operations include in response to aggregating the log data, determining deduplicated log data including one or more unique logs indicative of unique metrics associated with the computing system. The example operations include transmitting the deduplicated log data to a storage system.

[0014] In some implementations, the log data includes at least one of: (i) a higher throughput than the deduplicated log data or (ii) a higher cardinality than the deduplicated log data.

[0015] In some implementations, the aggregation layer is provided as part of a network layer.

[0016] In some implementations, the log data is ephemerally received by the aggregation layer.

[0017] In some implementations, the operations include generating, accessing the deduplicated log data from the storage system. In some implementations, the operations include accessing the deduplicated log data from the storage system.

[0018] In some implementations, the one or more aggregation parameters comprises metadata associated with the plurality of logs.

[0019] In some implementations, the metadata includes at least one of: (i) a time stamp or (ii) a log type associated with respective logs of the plurality of logs.

[0020] In some implementations, the operations include determining the aggregation layer is unavailable to receive additional log data. In some implementations, the operations include, based on determining the aggregation layer is unavailable to receive the additional log data, transmitting the additional log data to the storage system.

[0021] In some implementations, determining the aggregation layer is unavailable to receive the additional log data includes determining one or more faults associated with the aggregation layer, wherein the one or more faults are indicative of a fault tolerance.

[0022] In another example aspect, the present disclosure provides for one or more example non-transitory computer-readable medium storing instructions that are executable to cause one or more processors to perform operations. The example operations include receiving, from one or more client devices, log data including a plurality of logs indicative of metrics associated with a computing system, wherein the log data is received by an aggregation layer. The example operations include aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to increase or decrease a level of deduplication. The example operations include in response to aggregating the log data, determining deduplicated log data including one or more unique logs indicative of unique metrics associated with the computing system. The example operations include transmitting the deduplicated log data to a storage system.

[0023] In some implementations, the log data includes at least one of: (i) a higher throughput than the deduplicated log data or (ii) a higher cardinality than the deduplicated log data.

[0024] Other example aspects of the present disclosure are directed to other systems, methods, apparatuses, tangible non-transitory computer-readable media, and devices for performing functions described herein. These and other features, aspects and advantages of various implementations will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate implementations of the present disclosure and, together with the description, serve to explain the related principles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Detailed discussion of embodiments directed to one of ordinary skill in the art are set forth in the specification, which makes reference to the appended figures, in which:

[0026] FIG. 1 depicts an example computing ecosystem according to example aspects of the

present disclosure.

[0027] FIG. 2 depicts an example data flow pipeline according to example aspects of the present disclosure.

[0028] FIG. 3 depicts an example data flow pipeline according to example aspects of the present disclosure.

[0029] FIG. 4 depicts an example system architecture according to example aspects of the present disclosure.

[0030] FIG. 5 depicts an example data flow pipeline according to example aspects of the present disclosure.

[0031] FIG. 6 depicts an example flowchart diagram of an example method according to example aspects of the present disclosure.

[0032] FIG. 7 depicts an example computing ecosystem according to example aspects of the present disclosure.

DETAILED DESCRIPTION

[0033] Generally, the present disclosure is directed to techniques for processing high cardinality data, reducing throughput, and decreasing the use of storage resources for computing systems. For example, techniques according to the present disclosure provide an improved log consumption and processing model using an aggregation layer to deduplicate high cardinality log data prior to storing the logs. The processes described herein help to reduce ingestion throughput and storage costs for a computing system. The system of the present disclosure can more effectively consume data streams which include logs associated with the computing system by directly aggregating duplicative logs on a network layer and deduplicating the logs without the need to save the raw log data.

[0034] For example, in a server-client architecture for an application, multiple clients may send and receive messages, data, or other types of information to and from servers that run, provide, or implement the application. The data can include high cardinality data, such as event logs which increases the throughput of traffic between the clients and the server for the same type of log. As described herein, high cardinality data refers to data that has a large number of unique values or entities (e.g., a common column or field that can have many possible values). The large number of unique values in log data may represent a significant level of repetition and very little diversity. For example, a log file may be generated for a particular event which occurs hundreds or thousands of times per second. The only variance in the log files may include the time stamp (e.g., distinct value) indicating the time at which the event occurred. Handling a large number of distinct values over time may require complex data structures and algorithms to maintain performance. Further, storing duplicative logs where only one value (e.g., a column, field, etc.) is different can sharply increase storage costs. For instance, the degree of duplication may be a single unique log compared to hundreds of duplicative logs. To address these problems, the technology of the present disclosure allows for logs to be aggregated at the network layer (e.g., where traffic from clients are received) and deduplicated prior to being stored. Deduplication at the network layer reduces the overall throughput of the computing system by avoiding the processing of duplicative logs. Moreover, storing only unique log data reduces storage costs by also avoiding the storage of the duplicative logs.

[0035] According to example embodiments of the present disclosure, the aggregation layer may include software configured to receive event data from one or more clients associated with the computing system. The aggregation layer may be standalone software deployed on top of the network layer or may be included as a configuration of the network layer itself. For instance, the aggregation layer may include a distributed processing engine which receives log data including a plurality of log files and analyzes the log data to determine log characteristics. The log characteristics may indicate that the log is duplicative of other log files within the log data. By way of example, log data may include a number of log files (e.g., sixty log files) associated with a client

device. The sixty log files may include event data indicating that a user (e.g., user associated with the client device) took an action (e.g., action_id). The processing engine of the aggregation layer may analyze the sixty log files and determine based on log characteristics that the actions (e.g., action_id) associated with the sixty log files include three distinct actions (e.g., action_id is distinct value) taken by the user. As such the sixty log files may be aggregated and reduced to only three log files (e.g., three distinct log files per action per user).

[0036] Aggregating the log data may include grouping or filtering unique log files until only unique log data remains. As such, log data may have a higher cardinality and create a higher throughput than aggregated unique log data. Duplicative logs may be indicated by metadata or encoded parameters which indicate a number of times a duplicate log was received, or the unique values associated with a particular group of log files. For instance, one or more aggregation parameters may be configured to control the level of aggregation. Aggregation parameters may divide an event data stream (e.g., stream of event logs from one or more clients) based on a parameter. An example aggregation parameter can include a parameter such as time. By way of example, an aggregation parameter may be configured to aggregate log data based on a common timestamp. For instance, log data including one hundred logs files across four unique timestamps may be aggregated and reduced to four log files based on the unique timestamps. While examples herein describe timestamps as an aggregation parameter, the present disclosure is not limited to such an embodiment. Aggregation parameters may include any constant field or metadata associated with log data.

[0037] Once the log data has been aggregated, unique log data may be generated and consumed by an event processing or event logging system to expose the unique log data to downstream systems. For example, the unique log data may be stored and accessed by downstream systems for analytics purposes, root causes analysis (RCA), troubleshooting, and the like. In some examples, the aggregation layer may be unavailable (e.g., due to network outages, security incidents, etc.). In such cases, where the aggregation layer becomes unavailable, the log data may be stored for later processing. For instance, logs stored while the aggregation layer is unavailable may be read from storage and processed to continuously reduce the storage costs for the computing system.

[0038] The technology of the present disclosure may provide several benefits and technical effects. For instance, the technology of the present disclosure decreases the overall throughput and use of computing resources of a computing system by reducing the amount of log data processed and stored. As such, the technology may increase the overall stability, performance, and reliability of the application system by limiting the probability of incidents or outages due to high throughput or increased resource utilization. The technology of the present disclosure may also help to increase the flexibility of application systems without impacting performance due to the discovery of new client event data streams. Moreover, by reducing the overall throughput of the system data and amount of data stored by the system, the technology of the present disclosure may reduce storage costs and application complexity by simplifying high cardinality data.

[0039] Reference now will be made in detail to embodiments, one or more example(s) of which are illustrated in the drawings. Each example is provided by way of explanation of the embodiments, not limitation of the present disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations may be made to the embodiments without departing from the scope of the present disclosure. For instance, features illustrated or described as part of one embodiment may be used with another embodiment to yield a still further embodiment. Thus, it is intended that aspects of the present disclosure cover such modifications and variations.

[0040] FIG. 1 depicts an example computing ecosystem according to example aspects of the present disclosure. The example computing ecosystem **100** may include clients **101** which generate log data and transmit the log data through a network layer **102** to a computing system **103**. The computing system **103** may include an ingestion system **104** to ingest the log data, a storage system **105** which stores the log data, and a query system **106** which allows analytical applications **112** to

query the log data.

[0041] With respect to examples as described herein, the system **100** may be implemented on a server, combination of servers, or a distributed set of computing devices which communicate over a network such as the Internet. For example, the system **100** may be distributed using one or more physical servers, virtual private servers, or cloud computing. In other examples, the system **100** may be implemented as a part of or in connection with the clients **101**, where, for example, a distributed application (e.g., computing system **103**) is architected using a client-server relationship including independent or dependent services (e.g., microservices, etc.) to communicate over one or more networks with application clients (e.g., clients **101**).

[0042] Clients **101** can include desktop clients, web browsing clients, application clients, and the like. The clients **101** may be running on one or more client devices (e.g., desktop, mobile devices, etc.) and facilitate communications with one or more servers of the computing system **103**. For instance, the clients **101** may include a client program (e.g., software) that consumes services provided by a server program (e.g., computing system **103**). The clients **101** may request data or other services (e.g., microservices, etc.) from the one or more servers of the computing system **103** by calling functions in the server application (e.g., computing system **103**) over one or more networks. The clients **101** may generate log data as a user interacts with the clients **101** on a client device and the clients **101** interact with the computing system **103**. For instance, log data may be generated based on user actions, requests from the clients **101**, client performance, network events, etc.

[0043] The log data may be transmitted through a network layer **102** along with requests (e.g., queries, API calls, etc.) to interact with the server program (e.g., computing system **103**). Example log data may include availability logs (e.g., logs that track system performance and availability, etc.), resource logs (e.g., logs that indicate information on connectivity issues), security logs (e.g., logs that contain information regarding suspicious network activity, etc.), or any other type of log data. In some implementations, the computing system **103** may enable logging for the clients **101**. Logging may enable the computing system **103** to capture one or more data streams (e.g., event log data) from the clients **101** and store the log data for analysis or further processing. In some implementations, a discovery computing system may be utilized to discover new clients **101** which will stream log data to the computing system **103**. An example discovery computing system is further described with reference to FIG. 4.

[0044] Log data transmitted from clients **101** may be received and routed by a network layer **102**. The network layer **102** handles the routing and sending of data between different networks on behalf of the computing system **103**. For instance, clients **101** may be running on client devices connected to the computing system **103** over one or more networks. By way of example, the network layer **102** may receive requests, log data, etc., from clients **101** in the form of data packets, assemble the data packets, and determine the route for the requests or log data to reach the desired service (e.g., microservice, services, etc.) or destination internally within the computing system **103**. The network layer **102** may utilize one or more network protocols such as IP, ICMP, etc. to facilitate the routing of data packets. Additionally, or alternatively, the network layer **102** may transmit data packets to clients **101** (e.g., in response to requests, etc.) or other computing systems over one or more networks. As such the network layer **102** may facilitate inbound and outbound communications between the computing system **103** and the clients **101** operating on one or more networks. In some implementations, the network layer **102** may be deployed to network infrastructure, including but not limited to routers, switches, and other network interfaces.

[0045] The computing system **103** may include software running on one or more servers. For instance, the computing system **103** may be implemented as a part of or in connection with services or microservices. By way of example, microservices may be deployed in a container (e.g., standalone software package for a software application) using a container service, or on VMs (virtual machines) within a shared network. A container service may be a cloud service that allows

developers to upload, organize, run, scale, manage, and stop containers using container-based virtualization to orchestrate respective actions of the microservices. In some examples, a VM may include virtual computing resources which are not limited to a physical computing device.

[0046] The computing system **103** may include one or more subsystems. For instance, the computing system **103** may include an ingestion system **104**. The ingestion system **104** may include software running on one or more servers of the computing system **103**. The ingestion system **104** may be implemented using a virtual private cloud (VPC), container service, or may be deployed within a public cloud environment. The ingestion system **104** may be configured to ingest (e.g., receive, process, consume, etc.) data streams including log data. For instance, the ingestion system **104** may include a logging platform **107**. In some implementations, the logging platform **107** may be implemented using a container service. The logging platform **107** may include a distributed event store and streaming service for event processing. A distributed event processing store and streaming service may generate streaming data pipelines to allow for transmission of data between multiple systems. By way of example, the logging platform **107** may generate one or more data stream pipelines to allow log data generated by clients to be transmitted (e.g., streamed) to a distributed data store **109** where the log data may be consumed for data analytics purposes.

[0047] As further described herein, the ingestion system **104** may ingest log data from clients **101** that have subscribed to the logging platform **107** (e.g., enabled logging). Data ingestion may include streaming various types of log data from multiple sources (e.g., clients **101**, etc.) and storing the log data in a data store to be accessed and analyzed by analytical applications **112** using a distributed data store **109**. Log data may include event logs, (e.g., logs that records network traffic and usage data), server logs (e.g., logs that track actions on a server), system logs (e.g., logs that occur within the operating system), or any type of log produced by a software application. In some examples, log data may include structured, semi-structured, and unstructured logs. Structured log data may include JSON (JavaScript Object Notation) logs. Semi-structured logs may include HTML (Hypertext Markup Language), XML (Extensible Markup Language), or any other markup languages. Unstructured log data may include large text files including strings which are intended for human reading.

[0048] The computing system **103** may include a storage system **105**. The storage system **105** may include one or more systems and may be implemented using a virtual private cloud (VPC), container service, or may be deployed within a public cloud environment. For example, the storage system **105** may include a metadata service **108**, a distributed data store **109**, and a distributed file system **110**.

[0049] The metadata service **108** may be a service in the form of software code that is used to define the names, data types, etc., for the columns of the distributed data store **109**. In some examples, the metadata service **108** is deployed in a container. In some examples, the metadata service **108** may be deployed within the distributed data store **109**. In other examples, the metadata service **108** may be deployed on a virtual machine. The metadata service **108** may implement a universal schema for the storage system **105**. For instance, the clients **101** may generate high cardinality (e.g., repetitive) and variance (e.g., diverse) log data. By way of example, the clients **101** may generate high cardinality log data including repetitive log files where a single attribute or field (e.g., timestamp, client type, etc.) are different. High cardinality data may increase storage utilization due to the highly repetitive logs in which one or fewer attributes are different. In some examples, variance log data may be difficult to store in a single data store due to the differences in structure or format. The metadata service **108** may implement a universal schema such that repetitive log data which has been deduplicated and variance log data may be uniformly stored within the storage system **105**. As such the storage system **105** may further distribute log data to secondary storage systems where the log data may be further processed or more accessible for downstream systems.

[0050] For example, an aggregation layer deployed near the network layer may receive log data

generated by clients **101** and prior to being received by the ingestion system **104**, deduplicate (e.g., aggregate) high cardinality log data. The aggregation layer may deduplicate the high cardinality log data and indicate the repetitive values as metadata or encoded values thereby reducing throughput for the computing system **103**. An example of an aggregation layer is further described with reference to FIG. 2.

[0051] In some examples, the metadata service **108** may include a schemaless implementation. For instance, the distributed datastore **109** may store high variance log data in a non-tabular format. For example, the metadata service **108** may define one or more customized views of the data. In some examples, the customized view may be based on the type of log data (e.g., structured, unstructured, etc.). In other examples, the customized view may be based on the index type of the log data (e.g., aggregated log data, etc.). The customized view may be based on a plurality of index types. As such the schemaless implementation may provide flexibility to store high variance or high cardinality logs.

[0052] The distributed data store **109** may be a storage system configured to receive and store data in a distributed manner. By way of example, the distributed data store **109** may include an online analytical processing data store (e.g., OLAP database). An online analytical processing data store may ingest log data from data streams (e.g., the logging platform **107**) or from batch data sources (e.g., the distributed file system **110**) and allow analytical applications **112** to deliver data analytics in real-time. In some examples, the distributed data store **109** may be implemented using a container service. For instance, the distributed data store **109** may be deployed across multiple clusters of servers or containers.

[0053] The storage system **105** may include a distributed file system **110**. A distributed file system **110** may be any computer network where data is stored on more than one node (e.g., virtual machine, cluster of virtual machines, servers, etc.). For instance, the distributed file system **110** may include various types of nodes for storing data and segments with differing requirements. In some examples, the distributed file system **110** may include software that manages files in a storage system (e.g., storage system **105**). Example distributed file systems **110** may include software that manages files stored in a data lake, data warehouse, etc. In some examples, the distributed file system **110** may include servers associated with the distributed data store **109**. For instance, the distributed file system **110** may include real-time servers which store real-time data (e.g., data instantly available upon creation or processing) that may be accessed by the distributed data store **109**.

[0054] The distributed file system **110** may be associated with the ingestion system **104** and the storage system **105**. For instance, the ingestion system **104** may share computing resources with the distributed file system **110** and the storage system **105**. In some examples, the distributed file system **110** may manage servers and data for the logging platform **107**. For instance, the logging platform **107** may consume a data stream including raw log data for later or offline processing. Later or offline processing may include aggregating log data at a time later than ingestion. For instance, in the event of a network disruption where log data cannot be aggregated at the network layer **102**, raw log data may be ingested by the logging platform **107** and may be stored in offline servers to be processed at a time when normal operations resume. An example of offline processing is further described with reference to FIG. 5.

[0055] As described herein, the distributed file system **110** may store raw log data. Raw log data may include log files or messages which have not been aggregated. For example, the distributed file system **110** may include raw data log servers for storing the raw log data. In some examples, the logging platform **107** may receive a data stream and determine based on key values, log types, etc. that the logs should not be partitioned and directly store the raw log data within the distributed file system **110**.

[0056] By way of example, the logging platform **107** may receive a data stream (e.g., log data), from a client **101**, determine the topic for the log data is associated with offline data and partition

the topic into offline segments. Offline segments may include data streams where the log data has not been deduplicated. In some examples, the logging platform **107** may determine the topic is associated with offline data based on not receiving previous queries of the same log type (e.g., key value, log type, etc.). In other examples, the logging platform **107** may be configured to determine offline data and partition offline segments in response to detecting a network event (e.g., outage, disruption, etc.).

[0057] The distributed file system **110** may include archive servers for storing archived segments. Archived segments may include data which has not been accessed or queried in a threshold time. For instance, the logging platform **107** may receive a data stream from clients **101** and partition the topic into segments across a cluster. In some examples, the segments may not be accessed for a threshold time and the segments may be transmitted to the distributed file system **110** and stored in an offline server rather than being made available in real-time via the distributed data store **109**. In some examples, the segments may be stored in an archive server, where the segments will be archived. Log data stored in archived segments may be aggregated log data or raw log data.

[0058] In some examples, the logging platform **107** may be configured archive a data stream associated with a topic. For instance, the topic may include a key value (e.g., log type) which is rarely accessed. In some examples, the logging platform **107** may receive the data stream with an archive key value and directly partition segments within archive servers of the distributed file system **110**.

[0059] The computing system **103** may include a query system **106** for receiving and facilitating queries from analytical applications **112**. The query system **106** may include one or more systems and may be implemented using a virtual private cloud (VPC), container service, or may be deployed within a public cloud environment. For example, the query system **106** may include a proxy **111** and may include one or more analytical applications **112**. The analytical applications **112** may include software used to analyze log data. In some examples, the analytical applications **112** may include self-service dashboards. For instance, the analytical applications **112** may be used to measure and improve the performance of business operations of the computing system **103** or provide business intelligence by analyzing log data produced by clients **101**. In some examples, analytical applications **112** may be associated with a computing device with a user interface. An example of analytical applications **112** (e.g. consumers) is further described with reference to FIG. 4.

[0060] The proxy **111** may broker connections between the storage system **105** and the query system **106**. The proxy **111** includes a server, cluster of servers, virtual machines, or a computing system which serves as an intermediary between a client (e.g., analytical applications **112**, consumers, etc.) and a server (e.g., distributed data store **109**, storage system **105**, etc.). In some examples, the proxy **111** may be an API proxy. An API proxy may include an API gateway. In some examples, the API proxy may decouple the analytical application **112** from the backend services (e.g., clients **101**, logging platform **107**, distributed data store **109**, etc.). In some examples, the proxy **111** may facilitate API calls (e.g., query for log data) between the analytical applications **112** and the storage system **105**. For instance, the proxy **111** may handle identity management and authentication for query requests to ensure the security of the log data.

[0061] By way of example, the analytical applications **112** may receive a query request for log data. In some examples, the analytical applications **112** may automatically generate a query request (e.g., on page load of the analytical application **112**). The proxy **111** may receive the query and validate the user (e.g., user associated with the analytical application **112**) and ensure that the analytical application **112** (e.g., user authentication token, etc.) is authorized to access the data included in the query. In some examples, the proxy **111** may validate the identity of the user and authorize the access to the data by comparing user credentials or an API token to a security database. In some examples, the proxy **111** may validate the identity of the user and authorize the access to the data based on the classification of the log data. For instance, the log data may be public (e.g., view all

access, etc.) and the proxy **111** may authorize access to the data. The proxy **111** may transmit the query to the distributed data store **109** where the query results may be aggregated from the logging platform **107** and the distributed file system **110**. In some examples, the distributed data store **109** may transmit the query result through the proxy **111** to the analytical applications **112**. An example of the distributed data store **109** facilitating query requests is further described with reference to FIG. 4.

[0062] In some example, the query system **106** may automatically request data from the storage system **105**. For instance, the query system **106** may be scheduled to refresh data in the analytical application **112** on a specified cadence. In some examples, the query system **106** may request data on page load of the analytical application **112**. In other examples, the query system **106** may request data when data reflected in the analytical application **112** has reached a threshold age. For example, when data for an analytical application **112** has not been refreshed for **8** hours, the query system **106** may automatically request updated data.

[0063] FIG. 2 depicts an example dataflow pipeline according to example aspects of the present disclosure. The following description of dataflow pipeline **200** is described within an example implementation in which an aggregation layer **202** ephemerally receives log data from a plurality of clients **201A-C** and reduces a high traffic throughput by deduplicating the log data prior to ingestion.

[0064] The aggregation layer **202** can include software deployed near the network layer **102**. For instance, the aggregation layer **202** can include software deployed on top of the network layer **102** such that data (e.g., log data, requests, etc.) received by a computing system (e.g., computing system **103**) may first pass through the aggregation layer **202**. In some examples, the aggregation layer **202** can include software deployed directly behind the network layer **102**. In other examples, the aggregation layer **202** may be implemented as a function of the network layer **102** itself. The aggregation layer **202** may include software configured to execute one or more aggregation tasks (e.g., function, computing job, etc.) associated with one or more clients **201A-C**. For instance, respective clients **201A-C** may push log data directly to an aggregation task of the aggregation layer **202** that has been configured to aggregate log data for the specific client **201A-C**. An example of aggregation tasks is further described with reference to FIG. 3.

[0065] Log data produced by the plurality of clients **201A-C** may generate high throughput. High throughput may indicate that a network (e.g., network layer **102**) is transmitting or receiving a large amount of data per second. For instance, the plurality of clients **201A-C** may be generating large amounts of high cardinality log data and, due to logging being enabled, transmitting the large amounts of log data to the computing system **103**. High throughput can cause adverse impacts to the computing system **103** as more computing resources will be consumed to process and store the large amounts of log data.

[0066] As the plurality of clients **201A-C** transmit log data to the computing system **103**, the log data will first be ephemerally received by the aggregation layer **202** and processed prior to ingestion by the ingestion system **104**. Ephemerally receiving log data may include allowing the raw log data pushed (e.g., transmitted) from the plurality of clients **201A-C** to persist at the network layer **102** for a short period of time without storing the raw log data. By way of example, client **201A** and client **201B** may both generate thousands of logs per second indicating failed requests to a common service (e.g., service A) of the computing system **103**. The thousands of logs per second may include identical log files with variance among the timestamp of the failed request and the client identifier (e.g., client_id, etc.). The volume of logs files included in the log data may increase the throughput of traffic to the computing system **103** over time as additional computing resources will be required to store and process the log data. The aggregation layer **202** may receive the high throughput data streams (e.g., streams of log data) and based on one or more aggregation parameters aggregate the raw log data streams to produce deduplicated log data with a low throughput.

[0067] For instance, the aggregation layer **202** may be configured to deduplicate log data based on one or more predefined aggregation parameters. While examples herein describe predefined aggregation parameters, the present disclosure is not limited to such embodiment. The aggregation parameters may include dynamically determined aggregation parameters. For instance, the aggregation layer **202**, based on receiving log data over a threshold period of time may dynamically adjust or determine aggregation parameters based on highly duplicative log data.

[0068] Aggregation parameters can indicate one or more common log data attributes or features by which log data can be deduplicated. For example, the log data streams pushed to the computing system **103** from client **201A** and client **201B** may be deduplicated based on a “client id” attribute. In another example, aggregation parameters may include attributes or features such as log type. Failed requests for a common service (e.g., service A) may be associated with an error log type. As such, all error logs (e.g., log type) indicating the common service may be deduplicated based on the error log time for service A.

[0069] In yet another example, aggregation parameters may include attributes or features such as a timestamp. A timestamp indicates a day and time at which the log was generated for a particular event. Because timestamps may be a common attribute across all log data, log data can be deduplicated based on various times stamps (e.g., logs within a particular, hour, minute, second, etc.)

[0070] Log data received by the aggregation layer **202** may be analyzed to determine log attributes or features which satisfy the predetermined aggregation parameters and deduplicated based on the aggregation parameters. Deduplicating log data may include grouping log files of the log data which satisfy the aggregation parameter (e.g., timestamp, client identifier, etc.) and encoding the variance attributes or features. For instance, the variance data can include one or more unique fields or attributes associated with a plurality of log files.

[0071] By way of example, client **201B** and **201C** may each generate two thousand log files per minute indicating the same error message for service B of the computing system **103**. The only difference in the thousands of log files per minute may include the client_id (e.g., client **201B**, **20C**, etc.) and the timestamp associated with the log files. Based on an aggregation parameter such as a timestamp, the aggregation layer **202** may be configured to aggregate (e.g., deduplicate) the thousands of log files per minute. For instance, the aggregation layer **202** may group together all log files pushed from the client **201B** and **201C** over a one minute span (e.g., timestamp aggregation parameter) and encode the client_id (e.g., client **201B** and client **201C**). The result may generate deduplicated log data where one log file may represent each of the two thousand log files pushed from the respective client **201B** and **201C** over the one minute span. As such, high throughput traffic pushed from client **201B** and **201C** may be reduced at the aggregation layer **202** (e.g., network layer **102**) and a deduplicated log data may be transmitted to the computing system **103** as a low throughput data stream.

[0072] In some implementations, aggregation parameters may be used to increase or decrease a level of deduplication by the aggregation layer **202**. For instance, the aggregation parameters may be configured such that more or less log data is grouped together. By way of example, a timestamp aggregation parameter can be used to increase or decrease a level of deduplication. For instance, a timestamp aggregation parameter indicating a five minute time interval (e.g., group log files based on timestamps within five minutes) may increase deduplication by grouping more log files in a larger time interval as opposed to a two or one minute time interval. For example, a data stream that includes ten thousand log files per minute will generate more deduplicated log files with a smaller timestamp aggregation parameter than a larger timestamp aggregation parameter.

[0073] The computing system **103** may receive the deduplicated log data and utilize the ingestion system **104** to process and store the log data. The deduplicated log data may be much smaller in size than raw log data received by the aggregation layer **202**. For instance, the aggregation layer **202** may serve as a log processor and filter. The aggregation layer **202** may receive raw log data

pushed from clients **201A-C** and generate deduplicated log data with a lower throughput. The deduplicated log data may also be much smaller in size than the raw log data due to the removal of duplicate log files. As such, the deduplicated log data may be stored in the storage system **105** using less storage capacity.

[0074] FIG. **3** depicts an example dataflow pipeline according to example aspects of the present disclosure. The example dataflow pipeline **300** is described with an example implementation in which a discovery system **303** is utilized to concatenate data streams pushed from clients **201A-C** with one or more aggregation tasks **304A-C** of an aggregation layer **202**. The aggregation layer **202** may utilize the discovery system **303** to discover new clients **201A-C** or other computing systems which intend to stream log data to the computing system **103**.

[0075] For instance, the clients **201A-C** can include client application code **301** which enables the clients **201A-C** to render an application interface (e.g., on client device), execute application tasks, communicate (e.g., authenticate, send/receive requests, etc.), etc., with the computing system **103** over one or more networks. The clients **201A-C** can also include a reporter system **302** to notify the discovery system **303** of an intent to stream log data (e.g., logging enabled). The reporter system **302** may include software configured to monitor the software (e.g., client application code **301**) of the clients **201A-C** and serve as a client logger. A client logger may be used to troubleshoot and debug software issues, as well as to gather data for analysis and performance optimization. A client logger may also be used to monitor the usage and behavior of client software by end users. The reporter system **302**, once logging as been enabled, may be used to look up one or more assigned aggregation tasks **304A-C** within the aggregation layer **202** to which log data generated by the client **201A-C** should be pushed.

[0076] The discovery system **303** may include software running on one or more servers. For instance, the discovery system **303** may be implemented as a part of or in connection with services or microservices associated with the computing system **103**. By way of example, the discovery system **303** may be deployed in a container (e.g., standalone software package for a software application) using a container service, or on VMs (virtual machines) within a shared network. A container service may be a cloud service that allows developers to upload, organize, run, scale, manage, and stop containers using container-based virtualization to orchestrate respective actions of the microservices. In some examples, a VM may include virtual computing resources which are not limited to a physical computing device.

[0077] The discovery system **303** may be configured to track aggregation tasks **304A-C** that are available to receive a data stream from one or more clients **201A-C**. For instance, the discovery system **303** may serve as a broker between the clients **201A-C** and the aggregation layer **202**. Because the aggregation layer is deployed near the network layer **102**, aggregation tasks **304A-C** may be ephemeral in nature. For example, aggregation tasks **304A-C** may include software running in one or more containers which run for a period of time (e.g., to perform one or more aggregation tasks) and terminate. As such, aggregation tasks **304A-C** may be dynamically created based on the number of clients **201A-C** or other computing systems transmitting log data to the computing system **103**.

[0078] By way of example, the plurality of clients **201A-C** may respectively generate log data. For instance, one or more users may each interact with the plurality of clients **201A-C** via a client device. The client application code **301**, in response to the user's interaction may generate a series of requests, log files, etc. The reporter system **302** may log the activity of the client application code **301** and notify the discovery system **303** of an intent to stream log data to the computing system **103**. The discovery system **303** may indicate that aggregation tasks **304A-C** are available to receive the log data and the plurality of clients **201A-C** may push the log data directly to the aggregation layer **202** (e.g., aggregation tasks **304A-C**) where the log data may be deduplicated at the network layer **102** prior to being ingested and stored by the computing system **103**.

[0079] In some examples, the aggregation tasks **304A-C** may query the plurality of clients **201A-C**.

For instance, the discovery system **303** may store a registry of clients (e.g., of the plurality of clients **201A-C**) which have logging enabled. In the event of network disruption or an error (e.g., aggregation tasks **304A-C** fail to run) the aggregation layer **202** may query (e.g., discover) clients **201A-C** which have transmitted log data during the period of disruption or error. In some examples, the aggregation layer **202** (aggregation tasks **304A-C**) may process log data previously transmitted to the computing system during the period of disruption. An example of processing log data pushed from the clients **201A-C** during a period of disruption is further described with reference to FIG. 5.

[0080] FIG. 4 depicts an example dataflow pipeline according to example aspects of the present disclosure. The example dataflow pipeline **400** is described with an example implementation in which one or more consumers **402A-C** associated with an analytical application **112** retrieve data from the storage system **105**. The consumers **402A-C** may include software used to analyze log data stored in the storage system **105** using the analytical application **112**. For instance, the analytical application **112** may be a centralized application running on one or more servers of the query system **106**. The consumers **402A-C** may include self-service dashboards, analytical graphs, interactive charts, etc. which consume data and analytical information facilitated by the analytical application **112**. For instance, the consumers **402A-C** may be used to measure and improve the performance of business operations or provide business intelligence by analyzing data logs (e.g., deduplicated log files) produced by clients (e.g., clients **101**, clients **201A-C**, etc.).

[0081] Consumers **402A-C** may be associated with one or more computing devices. For instance, the computing device may be associated with a respective consumer **402A-C** which renders the analytical applications **112** via a user interface. The computing device may be any type of computing device, such as a smartphone, tablet computer, laptop computer, VR or AR headset device, and the like. As such, the computing device may include components such as a microphone, a camera, a satellite receiver, and a communication interface to communicate with external entities using any number of wireless communication protocols. In some examples, the computing device may store an instance of the consumers **402A-C** in a local memory to access the analytical application **112**. In some examples, the memory may store additional applications executable by one or more processors of the computing device, enabling access and interaction with one or more servers (e.g., query system **106**, storage system **105**, etc.) over one or more networks.

[0082] In some examples, the computing device may communicate with the query system **106** over one or more networks. By way of example, the analytical applications **112** may be stored on one or more servers of the query system **106** and enable consumer **402A-C** access. The computing devices may communicate via the consumers **402A** with the analytical application **112** by communicating over one or more networks (e.g., with the servers of the query system **106**). In some examples, the computing devices **407** may display the analytical applications **112** on the user interface of the computing device. For instance, the computing devices **407** may access and display the analytical applications **112** over a network such as the internet.

[0083] In some examples, the computing devices may be associated with one or more users. For instance, a user may interact with the consumers **402A-C** and display the analytical application **112** on the user interface of the computing device. In some examples, the user may submit a search query for log data stored in the storage system **105**. For instance, using the consumers **402A-C**, a user may access the analytical application **112** and may generate a search query. The query may return the requested log data from the storage system **105** and apply any defined analytical functions to generate a view via the user interface.

[0084] In other examples, the analytical applications **112** automatically generate a search query and expose the results to the consumers **402A-C**. For instance, upon accessing (e.g., page load) the consumers **402A-C**, the analytical applications **112** may generate a search query to retrieve log data needed to populate one or more user interface elements (e.g., analytical graphs, charts, dashboards, etc.) of the consumers **402A-C**. While examples herein describe consumers **402A-C** in the context

of performing data analytics, the present disclosure is not limited to such embodiment. Consumers **402A-C** may access stored log data for other purposes including, but not limited to troubleshooting, root-cause-analysis (RCA), trend monitoring, regulatory compliance, etc.

[0085] FIG. 5 depicts an example data flow pipeline according to example aspects of the present disclosure. The example dataflow pipeline **500** is described with an example implementation in which a contingency layer **501** checks to determine whether the aggregation layer **202** is available prior to pushing log data from clients **201A-C** to the aggregation layer **202**. The example contingency layer **501** may serve as a contingency function in the event of a period or partial period of disruption for the aggregation layer **202**, network layer **102**, computing system **103**, etc.

[0086] The contingency layer **501** may include software deployed near the network layer **102**. For instance, the contingency layer **501** may be deployed on top of the aggregation layer **202** or as a function of the aggregation layer **202**. In some examples, the contingency layer **501** may be integrated as a function of the network layer **102**. In other examples, the contingency layer **501** may be deployed in a container as a stand along function. For instance, the contingency layer **501** may be triggered as a standard function which must resolve as true (e.g., aggregation layer **202** available, etc.) before log data can be pushed to the aggregation layer **202**. In the event the contingency layer **501** determines the aggregation layer **202** is unavailable, the contingency layer **501** may be configured to bypass the aggregation layer **202** and route the raw log data streams directly to the ingestion system **104** to be processed and stored in the storage system **105**. The raw log data streams may include a higher throughput for the computing system **103** and increase storage utilization due to its high cardinality and duplicative characteristics.

[0087] By way of example, the plurality of clients **201A-C** may generate log data and attempt to push the log data to the computing system **103** for processing and storage. The contingency layer **501** may be configured to detect whether the aggregation layer **202** (e.g., aggregation computing tasks **304A-C**) are available to receive the log data streams. For instance, the contingency layer **501** may access the discovery system **303** to determine whether aggregation tasks **403A-C** are available. In some implementations, no aggregation tasks **403A-C** may be available, which can be indicative of a complete disruption period. In other implementations, the number of aggregation tasks **403A-C** may be available to aggregate log data may be less than what is required to aggregate the volume and number of data streams intending to push log data to the computing system **103** (e.g., partial disruption period.). Once determining a period or partial period of disruption, the contingency layer **501** may route the raw log data streams directly to the ingestion system **104**.

[0088] The contingency layer **501** may periodically check (e.g., query, poll, etc.) the discovery system **303** to determine whether a period or partial period of disruption exists. In some implementations, the contingency layer **501** may detect network related issues and based on the network related issues, determine a period or partial period of disruption. For instance, the contingency layer **501** may be implemented as a function of the network layer **102** and analyze network metrics such as throughput, traffic, and errors within the network layer.

[0089] Once the contingency layer **501** detects that a period or partial period of disruption has ended, the contingency layer **501** may resume pushing log data from the plurality of clients **201A-C** to the aggregation layer **202** where they may be aggregated prior to transmitting to the ingestion system **104**. For example, the contingency layer **501** may determine, based on the broker registry of the discovery system **303** that aggregation tasks **304A-C** are available to receive data streams from the plurality of clients **201A-C**.

[0090] Raw log data streams which have been ingested and stored in the computing system **103** may be processed offline by an instance of the aggregation layer **202**. For instance, the aggregation layer **202** may be configured to determine data streams that were directly streamed to the ingestion system **104** during the period or partial period of disruption. For instance, the contingency layer **501** may indicate these log data streams based on error logs pushed during the period or partial period, metrics indicating higher throughput during the period or partial period, etc. Once raw log

data has been identified, the aggregation layer can be applied (e.g., in a container, etc.) to reduce the storage capacity of the storage system **105**.

[0091] FIG. **6** depicts a flowchart diagram of an example method according to example aspects of the present disclosure. One or more portion(s) of the method **600** may be implemented by one or more computing devices such as, for example, the computing devices/systems described in FIGS. **2, 3, 4, 5**, etc. Moreover, one or more portion(s) of the method **600** may be implemented as an algorithm on the hardware components of the device(s) described herein. For example, a computing system may include one or more processors and one or more non-transitory, computer-readable media storing instructions that are executable by the one or more processors to cause the computing system to perform operations, the operations including one or more of the operations/portions of method **600**. FIG. **6** depicts elements performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that the elements of any of the methods discussed herein can be adapted, rearranged, expanded, omitted, combined, or modified in various ways without deviating from the scope of the present disclosure.

[0092] In an embodiment, the method **600** may include a step **602** or otherwise begin by receiving, from one or more client devices, log data including a plurality of logs indicative of metrics associated with a computing system, wherein the log data is published directly to an aggregation layer. For instance, clients (e.g., clients **101**, clients **201A-C**, etc.) may be included in a client-server relationship where independent or dependent services (e.g., microservices, etc.) of an application computing system (e.g., application server) communicate over one or more networks with application clients (e.g., clients **101**, clients **201A-C**, etc.).

[0093] The clients can include desktop clients, web browsing clients, application clients, and the like. The clients **101** may be running on one or more client devices (e.g., desktop, mobile devices, etc.) and facilitate communications with one or more servers of the computing system **103**. For instance, the clients **101** may include a client program (e.g., software) that consumes services provided by a server program (e.g., computing system **103**). The clients **101** may request data or other services (e.g., microservices, etc.) from the one or more servers of the computing system **103** by calling functions in the server application (e.g., computing system **103**) over one or more networks. The clients **101** may generate log data as a user interacts with the clients **101** on a client device and the clients **101** interact with the computing system **103**. For instance, log data may be generated based on user actions, requests from the clients **101**, client performance, network events, etc.

[0094] In an embodiment, the method **600** may include a step **604** or otherwise continue by aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to increase or decrease a level of deduplication. For instance, an aggregation layer **202** may ephemerally receive log data from a plurality of clients **201A-C**. The aggregation layer may include one or more aggregation tasks **304A-C** configured to process log data streams from one or more clients (e.g., clients **101**, clients **201A-C**, etc.). The one or more aggregation tasks **304A-C** may be configured to deduplicate log data streams based on one or more aggregation parameters.

[0095] By way of example, aggregation tasks **304A-C** may be configured to aggregate log data based on a timestamp aggregation parameter. The timestamp aggregation parameter may group together log files within the log data which share a common time stamp or have a timestamp within a predetermined or dynamic duration. For instance, aggregation tasks **304A-C**, may be configured to group together error log messages which have a common timestamps within a one minute interval. In some implementations, and aggregation tasks **304A-C** may be configured to group together error log messages which have common timestamps within a ten second interval.

[0096] The timestamp aggregation parameter may increase or decrease the level of deduplication of the log files. For instance, data streams which include high cardinality log data at a rate of fifty

thousand events per second may have an increased level of deduplication if a timestamp aggregation parameter associated with a one minute interval were applied due to the increased number of duplicative logs which are grouped together (e.g., during one minute) as opposed to a ten second time interval where a fewer number of log files may be grouped together (e.g., during ten seconds). The aggregation parameter may be adjusted and configured based on the client (e.g., client_id, etc.), log type (e.g., error event, security event, etc.), or any other log data characteristic. [0097] In an embodiment, the method **600** may include a step **606** or otherwise continue by, in response to aggregating the log data, generating deduplicated log data comprising a plurality of unique logs indicative of unique metrics associated with the computing system. For instance, the aggregation layer **202** may group together all log files pushed from the clients over a one minute span, ten second span, etc. (e.g., timestamp aggregation parameter) and encode the variant data (e.g., client_id, log type, etc.). The result may generate deduplicated log data where fewer log files may represent the thousands of raw log files pushed from the respective clients over the specified duration or timestamp. The deduplicated log data may be smaller in size than the raw log data and may lower the traffic throughput for the computing system **103**. For instance, raw log data (high cardinality, duplicative, etc.) can cause high throughput traffic when pushed from client **201B** and **201C**. The aggregation layer **202** (e.g., network layer **102**) may reduce the high throughput by deduplicating the log data and generating a low throughput data stream including the deduplicated log data (e.g., fewer number of log files) which will be ingested by the computing system **103**. Deduplicated log data may be transmitted to the computing system **103** as a low throughput data stream.

[0098] In an embodiment, the method **600** may include a step **608** or otherwise continue by transmitting the deduplicated log data to a storage system. For instance, the aggregation layer **202** may transmit the deduplicated log data to an ingestion system **104** where the log data may be processed and stored. Data ingestion may include streaming various types of log data from multiple sources (e.g., clients **101**, aggregation layer **202** etc.) and storing the log data in a data store where it may be accessed and analyzed by analytical applications **112**.

[0099] FIG. 7 depicts a block diagram of an example system **700** for implementing systems and methods according to example embodiments of the present disclosure. The example system **700** illustrated in FIG. 7 is provided as an example only. The components, systems, connections, and/or other aspects illustrated in FIG. 7 are optional and are provided as examples of what is possible, but not required, to implement the present disclosure. The example system **700** can include a client computing system **705** (e.g., clients **101**, **201A-C**, client devices, etc.). The example system **700** can include a server computing system **702** (e.g., computing system **103**, etc.). The example system **700** can include an analytics computing system **701** (e.g., consumers **402A-C**, analytical applications **112**, etc.). One or more of the client computing system **705**, the server computing system **702**, or the analytics computing system **701** can be communicatively coupled to one another over one or more communication network(s) **755**. The networks **755** can correspond to any of the networks described herein.

[0100] The computing device(s) **710** of the client computing system **705** can include processor(s) **715** and a memory **720**. The one or more processors **715** can be any suitable processing device (e.g., a processor core, a microprocessor, an ASIC, a FPGA, a controller, a microcontroller, etc.) and can be one processor or a plurality of processors that are operatively connected. The memory **720** can include one or more non-transitory computer-readable storage media, such as RAM, ROM, EEPROM, EPROM, one or more memory devices, flash memory devices, data registrar, etc., and combinations thereof.

[0101] The memory **720** can store information that can be accessed by the one or more

[0102] processors **715**. For example, the memory **720** (e.g., one or more non-transitory computer-readable storage mediums, memory devices, etc.) can include computer-readable instructions **1330A** that can be executed by the one or more processors **715**. The instructions **730** can be

software written in any suitable programming language or can be implemented in hardware. Additionally, or alternatively, the instructions **730** can be executed in logically and/or virtually separate threads on processor(s) **715**.

[0103] For example, the memory **720** can store instructions **730** that when executed by the one or more processors **715** cause the one or more processors **715** (e.g., of the client computing system **705**, etc.) to perform operations such as any of the operations and functions of the computing system(s) (e.g., operations computing system, etc.) described herein (or for which the system(s) are configured), one or more of the operations and functions for communicating between the computing systems, one or more portions/operations of method **600**, and/or one or more of the other operations and functions of the computing systems described herein.

[0104] The memory **720** can store processors **715** that can be obtained (e.g., acquired, received, retrieved, accessed, created, stored, etc.). The data **725** can include, for example, any of the data/information described herein. In some implementations, the computing device(s) **710** can obtain data from one or more memories that are remote from the client computing system **705**.

[0105] The computing device(s) **705** can also include a communication interface **740** used to communicate with one or more other system(s) remote from the client computing system **705**, such as server computing system **702**, and/or the analytics computing system **701**. The communication interface **740** can include any circuits, components, software, etc. for communicating via one or more networks (e.g., network(s) **755**, etc.). The communication interface **740** can include, for example, one or more of a communications controller, receiver, transceiver, transmitter, port, conductors, software and/or hardware for communicating data.

[0106] The server computing system **702** can include one or more computing device(s) **704** that are remote from the client computing system **705** and the analytics computing system **701**. The computing device(s) **704** can include one or more processors **707** and a memory **714**. The one or more processors **707** can be any suitable processing device (e.g., a processor core, a microprocessor, an ASIC, a FPGA, a controller, a microcontroller, etc.) and can be one processor or a plurality of processors that are operatively connected. The memory **714** can include one or more tangible, non-transitory computer-readable storage media, such as RAM, ROM, EEPROM, EPROM, one or more memory devices, flash memory devices, data registrar, etc., and combinations thereof.

[0107] The memory **714** can store information that can be accessed by the one or more processors **707**. For example, the memory **714** (e.g., one or more tangible, non-transitory computer-readable storage media, one or more memory devices, etc.) can include computer-readable instructions **722** that can be executed by the one or more processors **707**. The instructions **722** can be software written in any suitable programming language or can be implemented in hardware. Additionally, or alternatively, the instructions **722** can be executed in logically and/or virtually separate threads on processor(s) **707**.

[0108] For example, the memory **714** can store instructions **722** that when executed by the one or more processors **707** cause the one or more processors **707** to perform operations such as any of the operations and functions of the computing system(s) (e.g., advertisement server, etc.) described herein (or for which the system(s) are configured), one or more of the operations and functions for communicating between computing systems, one or more portions/operations of method **600** and/or one or more of the other operations and functions of the computing systems described herein. The memory **714** can store data **716** that can be obtained. The data **716** can include, for example, any of the data/information described herein.

[0109] The computing device(s) **704** can also include a communication interface **732** used to communicate with one or more system(s) that are remote from the system **702**. The communication interface **732** can include any circuits, components, software, etc. for communicating via one or more networks (e.g., network(s) **755**, etc.). The communication interface **732** can include, for example, one or more of a communications controller, receiver, transceiver, transmitter, port,

conductors, software and/or hardware for communicating data.

[0110] The analytics computing system **701** can include one or more computing device(s) **703** that are remote from the client computing system **705** and the server computing system **702**. The computing device(s) **703** can include one or more processors **706** and a memory **709**. The one or more processors **706** can be any suitable processing device (e.g., a processor core, a microprocessor, an ASIC, a FPGA, a controller, a microcontroller, etc.) and can be one processor or a plurality of processors that are operatively connected. The memory **709** can include one or more tangible, non-transitory computer-readable storage media, such as RAM, ROM, EEPROM, EPROM, one or more memory devices, flash memory devices, data registrar, etc., and combinations thereof.

[0111] The memory **709** can store information that can be accessed by the one or more processors **706**. For example, the memory **709** (e.g., one or more tangible, non-transitory computer-readable storage media, one or more memory devices, etc.) can include computer-readable instructions **718** that can be executed by the one or more processors **706**. The instructions **715** can be software written in any suitable programming language or can be implemented in hardware. Additionally, or alternatively, the instructions **718** can be executed in logically and/or virtually separate threads on processor(s) **706**.

[0112] For example, the memory **709** can store instructions **718** that when executed by the one or more processors **706** cause the one or more processors **706** to perform operations such as any of the operations and functions of the computing system(s) (e.g., user devices, etc.) described herein (or for which the user device(s) are configured), one or more of the operations and functions for communicating between systems, one or more portions/operations of method **600** and/or one or more of the other operations and functions of the computing systems described herein. The memory **709** can store data **712** that can be obtained. The data **712** can include, for example, any of the data/information described herein.

[0113] The computing device(s) **703** can also include a communication interface **721** used to communicate computing device/system that is remote from the analytics computing system **701**, such as server computing system **702** or client computing system **705**. The communication interface **721** can include any circuits, components, software, etc. for communicating via one or more networks (e.g., network(s) **755**, etc.). The communication interface **721** can include, for example, one or more of a communications controller, receiver, transceiver, transmitter, port, conductors, software and/or hardware for communicating data.

[0114] The network(s) **755** can be any type of network or combination of networks that allows for communication between devices. In some implementations, the network(s) **755** can include one or more of a local area network, wide area network, the Internet, secure network, cellular network, mesh network, peer-to-peer communication link and/or some combination thereof and can include any number of wired or wireless links. Communication over the network(s) **755** can be accomplished, for example, via a communication interface using any type of protocol, protection scheme, encoding, format, packaging, etc.

[0115] Computing tasks discussed herein as being performed at certain computing device(s)/systems may instead be performed at another computing device/system, or vice versa.

[0116] Such configurations may be implemented without deviating from the scope of the present disclosure. The use of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. Computer-implemented operations may be performed on a single component or across multiple components. Computer-implemented tasks or operations may be performed sequentially or in parallel. Data and instructions may be stored in a single memory device or across multiple memory devices.

[0117] The technology discussed herein makes reference to servers, databases, software applications, and other computer-based systems, as well as actions taken, and information sent to and from such systems. The inherent flexibility of computer-based systems allows for a great

variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein may be implemented using a single device or component or multiple devices or components working in combination. Databases and applications may be implemented on a single system or distributed across multiple systems. Distributed components may operate sequentially or in parallel.

[0118] Aspects of the disclosure have been described in terms of illustrative implementations thereof. Numerous other implementations, modifications, or variations within the scope and spirit of the appended claims may occur to persons of ordinary skill in the art from a review of this disclosure. Any and all features in the following claims may be combined or rearranged in any way possible. Accordingly, the scope of the present disclosure is by way of example rather than by way of limitation, and the subject disclosure does not preclude inclusion of such modifications, variations or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art. Moreover, terms are described herein using lists of example elements joined by conjunctions such as “and,” “or,” “but,” etc. It should be understood that such conjunctions are provided for explanatory purposes only. The term “or” and “and/or” may be used interchangeably herein. Lists joined by a particular conjunction such as “or,” for example, may refer to “at least one of” or “any combination of” example elements listed therein, with “or” being understood as “and/or” unless otherwise indicated. Also, terms such as “based on” should be understood as “based at least in part on.”

[0119] Those of ordinary skill in the art, using the disclosures provided herein, will understand that the elements of any of the claims discussed herein may be adapted, rearranged, expanded, omitted, combined, or modified in various ways without deviating from the scope of the present disclosure. Some implementations are described with a reference numeral, for example illustrated purposes and are not meant to be limiting.

Claims

1. A computer-implemented method comprising: receiving, from one or more client devices, log data comprising a plurality of logs indicative of metrics associated with a computing system, wherein the log data is received by an aggregation layer; aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to increase or decrease a level of deduplication; in response to aggregating the log data, determining deduplicated log data comprising one or more unique logs indicative of unique metrics associated with the computing system; and transmitting the deduplicated log data to a storage system.
2. The computer-implemented method of claim 1, wherein the log data has at least one of: (i) a higher throughput than the deduplicated log data or (ii) a higher cardinality than the deduplicated log data.
3. The computer-implemented method of claim 1, wherein the aggregation layer is provided as part of a network layer.
4. The computer-implemented method of claim 1, wherein the log data is ephemerally received by the aggregation layer.
5. The computer-implemented method of claim 1, further comprising: accessing the deduplicated log data from the storage system; and computing analytics data based on the deduplicated log data, the analytics data being associated with the log data.
6. The computer-implemented method of claim 1, wherein the one or more aggregation parameters comprises metadata associated with the plurality of logs.
7. The computer-implemented method of claim 6, wherein the metadata comprises at least one of: (i) a time stamp or (ii) a log type associated with respective logs of the plurality of logs.
8. The computer-implemented method of claim 1, further comprising: determining the aggregation

layer is unavailable to receive additional log data; and based on determining the aggregation layer is unavailable to receive the additional log data, transmitting the additional log data to the storage system.

9. The computer-implemented method of claim 8, wherein determining the aggregation layer is unavailable to receive the additional log data comprises determining one or more faults associated with the aggregation layer, wherein the one or more faults are indicative of a fault tolerance.

10. A computing system comprising: one or more processors; and one or more non-transitory computer-readable medium storing instructions that are executable by the one or more processors to cause the computing system to perform operations, the operations comprising: receiving, from one or more client devices, log data comprising a plurality of logs indicative of metrics associated with the computing system, wherein the log data is received by an aggregation layer; aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to increase or decrease a level of deduplication; in response to aggregating the log data, determining deduplicated log data comprising one or more unique logs indicative of unique metrics associated with the computing system; and transmitting the deduplicated log data to a storage system.

11. The computing system of claim 10, wherein the log data has at least one of: (i) a higher throughput than the deduplicated log data or (ii) a higher cardinality than the deduplicated log data.

12. The computing system of claim 10, wherein the aggregation layer is provided as part of a network layer.

13. The computing system of claim 10, wherein the log data is ephemerally received by the aggregation layer.

14. The computing system of claim 10, wherein the operations further comprise: accessing the deduplicated log data from the storage system; and computing analytics data based on the deduplicated log data, the analytics data being associated with the log data.

15. The computing system of claim 10, wherein the one or more aggregation parameters comprises metadata associated with the plurality of logs.

16. The computing system of claim 15, wherein the metadata comprises at least one of: (i) a time stamp or (ii) a log type associated with respective logs of the plurality of logs.

17. The computing system of claim 10, wherein the operations further comprise: determining the aggregation layer is unavailable to receive additional log data; and based on determining the aggregation layer is unavailable to receive the additional log data, transmitting the additional log data to the storage system.

18. The computing system of claim 17, wherein determining the aggregation layer is unavailable to receive the additional log data comprises determining one or more faults associated with the aggregation layer, wherein the one or more faults are indicative of a fault tolerance.

19. A non-transitory computer-readable medium storing instructions that are executable by one or more processors to perform operations, the operations comprising: receiving, from one or more client devices, log data comprising a plurality of logs indicative of metrics associated with a computing system, wherein the log data is received by an aggregation layer; aggregating the log data by deduplicating the plurality of logs using one or more aggregation parameters, wherein the one or more aggregation parameters are configurable to increase or decrease a level of deduplication; in response to aggregating the log data, determining deduplicated log data comprising one or more unique logs indicative of unique metrics associated with the computing system; and transmitting the deduplicated log data to a storage system.

20. The non-transitory computer-readable medium of claim 19, wherein the log data has at least one of: (i) a higher throughput than the deduplicated log data or (ii) a higher cardinality than the deduplicated log data.
